

APS110 – Lecture 23

See hand written notes for drawings.

Polymers – time dependent deformation

1. Apply fixed load → observe strain increasing with time
 2. Apply a fixed strain → observe stress relax over time (decreasing stress)
 - Ex. guitar → as the strings stay in place over a long time, they stretch and you have to retune (particularly noticeable when new strings are placed)
- Viscous substance → apply force over time
 - When the force is applied, there is no instantaneous deformation
 - There will be a period of time-dependent deformation (curve slope $\neq 0$) then the load will be released and the viscous material will remain in its final shape
 - No elastic recovery
 - Polymers are somewhere between these two extremes – we call them **VISCOELASTIC**
 - They behave a little like both categories of materials
 - When the load is applied, there is some instantaneous deformation, followed by a period of time-dependent deformation
 - Then, when the load is released, there is some elastic behaviour that brings it back part way to its original shape
 - **ALL polymers are viscoelastic to some extent**
 - So, we have stress relaxation over time → quantify with **relaxation modulus**
 - We impose a fixed strain and we're recording what the stress does over time
 - With polymers we know the stress will decrease

Plot $\log E_R$ vs. Temp.

- There seem to be two temperature points where the curve changes dramatically → the molecules have enough kinetic energy to overcome interaction strength
- At the **lower** critical temperature, the kinetic energy gets high enough to allow molecules to overcome SECONDARY bonds/interactions in the **AMORPHOUS region** of the material
 - **T_g = glass transition temperature**
 - Easier to separate chains in the amorphous region (compared to crystalline region) thus it takes a lower temperature to give the molecules enough energy to break free of these bonds
- At the **higher** critical temperature, the kinetic energy gets high enough to allow molecules to overcome secondary bonds/interactions in the **CRYSTALLINE region** of the material
 - **T_m = melting**
 - The crystalline region more packed, thus is harder to overcome secondary bonding between chains

Plexiglass® – polymethylmethacrylate (PMMA)

- *Acrylate*, tells us it's an acrylic
- Has a LARGE functional group
 - This prevents PMMA from crystallizing
- PMMA is *amorphous*, which is significant to most of its applications
 - Light can pass through amorphous polymers without refraction/scattering
 - This means PMMA is **transparent**
 - In a semi-crystalline polymer the crystal boundaries will scatter the light (as they have a different refraction index)
- There is a relationship between the structure of the polymer and its optical properties
 - This is not the only thing that affects the optical properties, as you can imagine a completely crystalline metal is not transparent
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